CSA1718 : Artificial Intelligence

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1.8 Puzzle Problem:

```
from heapq import heappush, heappop

start = (1, 2, 3, 4, 0, 6, 7, 5, 8)
goal = (1, 2, 3, 4, 5, 6, 7, 8, 0)

def h(state):
    return sum(state[i] != 0 and state[i] != goal[i]
               for i in range(9))

pq = [(h(start), 0, start, [])]
visited = set()

while pq:
    f, cost, state, path = heappop(pq)
```

```
if state == goal:
    for s in path + [state]:
        print(s[:3])
        print(s[3:6])
        print(s[6:])
        print()
    break

if state in visited:
    continue
visited.add(state)

z = state.index(0)
r, c = divmod(z, 3)

for dr, dc in [(1,0), (-1,0), (0,1), (0,-1)]:
    nr, nc = r + dr, c + dc
    if 0 <= nr < 3 and 0 <= nc < 3:
        j = nr * 3 + nc
```

```
            new = list(state)
            new[z], new[j] = new[j], new[z]
            new = tuple(new)

            if new not in visited:
                heappush(pq, (cost + 1 + h(new),
                              cost + 1, new, path + [state]))
```

Output:

```
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```